

GPlot V0.90 Command Cheat Sheet

SYSTEM	
DEVICE	NAME [FILE [DIMENSIONS]] - Select output device [W,H(+X+Y)]
CLEAR	[FILE] - Clear drawing / and output file, change output file name.
FILE	File screen with colour (device dependent).
OKEY	NAME [PARAMETER-LIST] - Read commands from file (nestable)
HELP	TOPIC - Display help on topic
STATUS	Display status information
MEMTEST	Test dynamic memory. Generate graph plotting test data
GET	NAME - Get an indirect permanent file (ignored if not on NOS)
LOGFILE	NAME - Open a command log file
MAXPOINTS	NUMBER - Set max data points in internal arrays (default 1000)
NSTACK	N - Set evaluator stack size (minimum 4, default 8)
VERSION	[N] - Version information. Print or put in string register N
WAIT	For user response (enter) on interactive device (eps/svg ignored)
RESET	Reset most state to defaults. Not device or read data.
PREFIX	PATH - Set a prefix to prepend to all filenames. ignored on NOS.
EXIT	Exit GPlot

DEVICES	
GTERRA	Python/Qt/OpenGL based colour graphics terminal
TEK4K	Textronix 4010/4014 terminal
EPSCOL	NAME [W,H(+X+Y)] - Encapsulated PostScript file (W,H,X,Y INCH)
	default size 5.5-8.0+5.0 inches
SVG	NAME [W,H] - Scalable Vector Graphics file (W,H PIXEL)
	default size 800,800 pixels

COLOURS AND STYLES	
CSGROUP	ALL/GENERAL/TEXT/ANNOT - Colour/style group to set
COLOUR	R G B - Set rgb colour to use, range 0 to 1
WIDTH	NUMBER - Set line width
STYLE	STYLE SOLID/DASH/DOT/DASHDOT [LNT] - Line style, dash len (all groups)

GRAPH PLOTTING	
READ	NAME XCOL YCOL [YECOL [XECOL]] - Read a data file using columns. ... NAME=HERE, use command input to EOF line ... XCOL YCOL etc, are column numbers (1 ...) space or comma separated
XYPOINT	Draw xy graph with points
XYLINE	Draw xy graph with lines
XYHISTOGRAM	Draw xy histogram
GRMOVE	X Y - Move to graph coords (X,Y)
GRDRAW	X Y - Draw to graph coords (X,Y)
XVALTO	Find both axis ranges automatically
XRANGE	XLO YHI - Set X axis range
YRANGE	YLO YHI - Set Y axis range
XYSCALE	Keep previous X,Y axis ranges
XYLINEAR	Use linear X axis
XYLINEAR	Use linear Y axis
XLOG	Use log X axis
YLOG	Use log Y axis
INTVALUES	NONE/X/Y/BOTH - Try to use integers for axis values, if possible.
GRID	NONE/X/Y/BOTH - Grid graph in axis
INTERPOLATE	LINEAR/CUBIC/QUINTIC N - Interpolate with N intermediate points
ASYMMETRIC	ON/OFF - Asymmetric Y error bars (off for X & Y error bars)
ABUT/ABUT+SHADE/LINES/WIDE+SHADE [WIDTH]	- Histogram style
ON/OFF	- Turn graph annotation on/off
ON/OFF	- Turn right edge annotation on/off (drawn only if annotate off)
TEXT [LOWER]	- Set title. If lower is yes, below instead of above
XLABEL	TEXT - Set X axis label
YLABEL	TEXT - Set Y axis label
RYLABEL	TEXT - Set right edge Y axis label
GSTYLE	BOXED/AXES/OPEN - Overall graph drawing style. (axes must be in range to draw, see ACUT)
ACUT	XO YO - Set point through which axes pass (default: (0,0))
MARKER	NUMBER - Set point marker number
USEKEY	Prepare to create a key (legend) for the graph
ADDKEY	TEXT - Add a key for the current graph line/points
KEYS	Draw the accumulated keys (after all lines/points)
GRAPHMODE	ON/OFF - Internally set bounds to same aspect ratio of the device. ... if not ON, the graph appears in a centered, square, subregion
SUBFIGGRID	NX NX IX IX [SHRINK] - Set up pane to draw next graph as a sub-figure. ... Figures arranged in NX X NY grid, this figure at IX,IY.

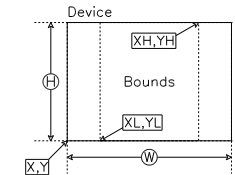
L-SYSTEM	
LSYSTEM	NRULES NITER ANGLE - Generate and draw an L-system.
	... Axiom in STRING 1, rules in STRINGS 2-9
	... Alphabet: F A B C D E M X Y + - []
F and D	Draw a unit length line segment at the current drawing angle.
M	Move by unit length at the current drawing angle.
+	Turn counter-clockwise by the turning angle.
-	Turn clockwise by the turning angle.
[	Push the current position and angle on to a stack.
]	Pop a position and angle off the stack and make them current.

BASIC DRAWING	
BOUNDS	XL XH YL YH - Set plot bounds (user world coordinate system)
PANE	XL XH YL YH - Set pane (clipping) area.
	... also area in which graph will be plotted.
CANVAS	XL XH YL YH - Set bounds and pane (to be the same)
UNBLANK	Stop using any pane
UNBLANK	XL XH YL YH - Set blank area
MOVE	Stop using any blank area
MOVE	PANE/BLANK/BOUNDS/DEVICE - Outline an area
MOVE	X Y - Move to position
MOVE	X Y - Draw to position
PATH	C/O - Draw a closed / open polyline.
	... coordinates from X,Y, args (see over).
CURVE	C/O - Draw a closed / open smooth curve.
BEZIER	C/O - Draw a closed / open bezier curve.
CIRCLE	X Y R - Draw circle, center X,Y, radius R
ARC	X Y R A1 A2 - Draw a circular arc, center X,Y, radius R
	... start angle A1, end angle A2
RECT	X Y W H - Draw a rectangle, bottom left at X,Y, wd W, ht H.
CRECT	X Y W H - Draw a rectangle, center X,Y, width W, height H.
	SET FONTNAME = Font/symbols/markers to use.
	... SET: 1 2 3 S M for 3 alphabets, symbol, marker.
	List available fonts
LISTFONT	HEIGHT - Text, symbol or marker height in user bounds units
SYMHGT	ANGLE - Text drawing angle wrt X axis (ccw decares)
SYMHGT	TEXT - Draw text (with format control)
CTEXT	WIDTH TEXT - Text horizontally centered on current XPOS,YPOS.
	... scaled to WIDTH using units (0 for no scaling, use SYMHGT)
MARKER	NUMBER YES - Draw marker at current position

HIGHER DRAWING - TEXT BOXES	
BOXTEXT	TEXT [X Y] - Draw text in a box with centre or bottom left
	... At X,Y or at the last incremented box position. Box
	... Position increments by specified deltas after each draw.
	... Text is centered vertically and centered or flush left
	... HORIZONTALLY, Text may have up to 5 lines separated by
	... BACKSLASHES.
BOXPSIZE	WIDTH HEIGHT BOTLEFT - Set box size. BOTLEFT YES/NO sets.
	... MEANING OF BOXTEXT X,Y NO = center, YES = bottom left.
BOXPHATCH	SPACING ANGLE MODE - Hatch spacing, angle (in spacings), mode:
	... NONE/VERT/HORIZ/BOTH
BOXPTEXT	WSCALE MODE PLOUSHLR - Set how text is drawn. MODE: FIX/SCALE.
	... WSCALE: LINEAR, LOG, 2, 1, RT, A2, 6, 2, 1
BOXPBOX	WIDTHFRAC: Width fraction of box width for text in SCALE mode.
BOXPDELTA	NONE/OUTER/INNER/BOTH - Draw the box outline, inner box around text.
	DX DY = Automatic box position step after every box draw.

HIGHER DRAWING - DECORATED LINES & LABELS	
LINE	LINESPEC - Draw a decorated polyline defined by LINESPEC
	... LINESPEC is a sequence of POINTSPECS separated by) characters.
	... POINTSPEC is (type)x,Y[A] where x and y are bounds coords.
	... (type) is P for a plain, unadorned, location. A for an
	... arrow with automatically determined direction, or S for a skip
	... to a shape to skip over on the line, or L for a loop.
	... (very short) arbitrary text to place in the label at the mid
	... of the line segment the POINTSPEC starts.
ARROWPARAM	TYPE SIZE SHARPNESS BARBENESS - Arrow parameters. TYPE:
	... OPEN/CLOSED/BARBED, Size in bounds units, SHARPNESS 0.5/5.
	BARBENESS fraction of arrow head in barb.
LINEPARAM	SKIPSKEAL ANNOTSCALE - Scale of skips and mid-line annotations.
GLABEL	GX GY LENGTH ANGLE TEXT - Draw a label with an arrow and boxed text.
	... GX,GY are graph coord coordinates to which the arrow points.
	... LENGTH is the length of the arrow line in bounds units, ANGLE
	... is the angle of the line wrt +X in degrees. Text is the
	... text to appear in the label box
ALABEL	X Y LENGTH ANGLE TEXT - As GLABEL but, X,Y in bounds coords.

EVALUATOR (see over for operators)	
ERANGE	BEGIN START STOP NCELEM - Set range of values in X/Y
	... stack register. BASE 1 linear, else log base.
EVAL	RPN [ARG] - Execute a procedure/evaluate a function using an RPN
	... NOTATION, ARG 1,2, etc. refer to proc registers using @
ITEVAL	START END STEP RPN - Iterated evaluation over start to end by step.
2DEVAL	RPN = As EVAL, but preserve both X and Y data (see over).
3DEVAL	START END STEP RPN = As ITEVAL, but preserve both X and Y.
PROC	N RPN - Store RPN code string in proc register N. Use EVAL @N
LOADPROC	N NAME - Load a named procedure from GPLPROC file into proc reg N
TEXT	TEXT - Set contents of string register N
STO	N X - Store real number X in memory register N
RCO	N - Display contents of memory register N
DI	Y = Set value for divide-by-zero trap. Default 1e-9, 0=no trap.
BBSTART	Accumulate bounding box instead of drawing with M & D operators.
BBEND	Exit bounding box mode and return to drawing with M & D operators.
BBSET	[SCALE] - Set bounds to match bounding box found with BBSTART/BBEND.



TEXT STRING MARKUP CODES	
..U	Go to upper case
..L	Start/end lower case
..B	Backspace
..+2+3	Select a font set
..+	Start/end superscript (2 levels)
..+..	Start/end subscript (2 levels)
..n	Reset sub/superscript level to 0
..n	Sub/sup fully below/above last char
..NUM \$ DEN \$	Construct a fraction
..nn	Output symbol with code nn
..nn	Output marker with code nn
..nn	Output alphabetic with code nn
..=	Start/end underlining

COMMAND LINE OPTIONS	
GLOTKEY=VAL	To run on NOS
UGLOTKEY=VAL	To run on Unix-like systems.
OBJ=FILE	Name of obj file to run.
PARAM=string	Parameters for obj file in dbl args
GET=YES or NO	Auto GET files on NOS
SAVE=YES or NO	Auto SAVE files on NOS
DEFL=YES or NO	Show expanded parameters
QUITE=YES or NO	Show expanded command
SLIDE=YES or NO	Adjust graph layout for larger text

PLNOTES MARKDOWN INTEGRATION	
{ GLOT COMMAND	
{ GLOT COMMAND	
... etc.	
... no DEVICE or CLEAR	
	(blank line ends GLOT commands)
Use:	
MODE=EXEC,BUILD,PLNOTES,ARG=ANOTE.	
... to create ANOTE.HTML	

GPlot V0.90 Evaluator Cheat Sheet

OPERANDS		
(DIGITS)	(-- C1)	SET TOS ARRAY TO A LITERAL CONSTANT.
X	(-- A1)	SET TOS TO X RANGE ARRAY.
PI	(-- A1)	SET TOS TO PI.
E	(-- C1)	SET TOS TO E.
I	(-- C1)	SET TOS TO ITERATION NUMBER FROM ITEMP. 1 IF IN EVAL.
IDX	(-- A1)	SET TOS TO THE ARRAY ELEMENT INDEX.
TPWPI	(-- C1)	SET TOS TO π .
PI/2	(-- C1)	SET TOS TO $\pi/2$.
STACK MANIPULATION		
SWAP or S	(A1 A2 -- A2 A1)	SWAP OR EXCHANGE TOP 2 STACK ARRAYS.
DUP or &	(A1 -- A1 A1)	DUPPLICATE TOP OF STACK
C	(AN ... C1 -- AN)	GET STACK LEVEL AT DEPTH C1 INTO TOP OF STACK
POP	(--)	POP TOP OF STACK
CL	(--)	CLEAR STACK
SETX	(A1 -- A1)	OVERWRITE RANGE, GRAPH X VALUES WITH STACK TOP VALUES: X = A1
SETY	(A1 -- A1)	OVERWRITE ORTH Y VALUES WITH STACK TOP VALUES: Y = A1
XLIN	(C1 C2 --)	X = C1 TO C2 IN C3 LINEAR STEPS
XLOG	(C1 C2 C3 C4 --)	X = C1 TO C2 IN C3 LOG STEPS
C1	(C1 C2 --)	C1 = A1*(C2)
IOU	(C1 C2 -- C1 * C2)	C1 = MOD(C1,C2), C2 = (C1/C2): 1D TO 2D INDEX CONVERT, 0 BASED.
ITU	(C1 C2 -- C1 / C2)	C1 = MOD(C1,C2)+1, C2 = (C1/C2)+1: 1D TO 2D INDEX CONV, 1 BASED.
BASIC ARITHMETIC		
+	(A1 A2 -- A1 + A2)	ADD ARRAY ELEMENTS: A1 = A1 + A2
-	(A1 A2 -- A1 - A2)	SUBTRACT ARRAY ELEMENTS: A1 = A1 - A2
R-	(A1 A2 -- A1 * A2)	REVERSE SUBTRACT: A1 = A2 - A1
+	(A1 A2 -- A1 * A2)	MULTIPLY ARRAY ELEMENTS: A1 = A1 * A2
**	(A1 A2 -- A1 ** A2)	EXPONENTIATION: A1 = A1 ** A2
/	(A1 A2 -- A1 / A2)	DIVIDE ARRAY ELEMENTS: A1 = A1 / A2
RCP	(A1 -- A1)	REVERSE DIVIDE: A1 = 1.0 / A1
CHS	(A1 -- A1)	RECIPROCAL: A1 = 1.0 / A1
ABS	(A1 -- A1)	A1 = ABS(A1)
MATH FUNCTIONS		
SIN	(A1 -- A1)	A1 = SIN(A1)
COS	(A1 -- A1)	A1 = COS(A1)
TAN	(A1 -- A1)	A1 = TAN(A1)
ASIN	(A1 -- A1)	A1 = ARCSIN(A1)
ACOS	(A1 -- A1)	A1 = ARCCOS(A1)
ATAN	(A1 -- A1)	A1 = ARCTAN(A1)
SINH	(A1 -- A1)	A1 = SINH(A1), DOMAIN [X] (-742.36
COSH	(A1 -- A1)	A1 = COSH(A1), DOMAIN [X] (-742.36
TANH	(A1 -- A1)	A1 = TANH(A1), DOMAIN [X] (-742.36
ASINH	(A1 -- A1)	A1 = ASINH(A1)
ACOSH	(A1 -- A1)	A1 = ACOSH(A1)
ATANH	(A1 -- A1)	A1 = ATANH(A1)
LOG	(A1 -- A1)	BASE E LOGARITHM: A1 = LN(A1)
LOG10	(A1 -- A1)	BASE 10 LOGARITHM: A1 = LOG10(A1)
LOG2	(A1 -- A1)	BASE 2 LOGARITHM: A1 = LOG2(A1)
EXP	(A1 -- A1)	A1 = E ** A1, DOMAIN: -675.81 TO 741.66
RAND	(A1 -- A1)	UNIFORMLY DISTRIBUTED RANDOMS (0..1): A1 = A1 * RAND
SEED	(C1 --)	SET RANDOM NUMBER SEED TO A1(I)
GCD	(C1 C2 -- C1)	FIND GREATEST COMMON DIVISOR OF C1, C2
NUMBER RANGE RELATED		
MIN	(A1 A2 -- A1)	A1 = MIN(A1,A2)
MAX	(A1 A2 -- A1)	A1 = MAX(A1,A2)
MOD	(A1 A2 -- A1)	A1 = MOD(A1,A2) OR A1 = A1 % A2
SIGN	(A1 -- A1)	A1 = (A1 > 0) ? 1 : -1
CONDITIONALS		
ODD	(A1 -- A1)	A1 = 1 WHERE INT(A1) IS ODD ELSE 0
LT	(A1 A2 -- A1 A2 A3)	A1 = (A1 < A2) ? 1 : 0
LE	(A1 A2 -- A1 A2 A3)	A3 = (A1 <= A2) ? 1 : 0
GT	(A1 A2 -- A1 A2 A3)	A3 = (A1 > A2) ? 1 : 0
GE	(A1 A2 -- A1 A2 A3)	A3 = (A1 >= A2) ? 1 : 0
EQ	(A1 A2 -- A1 A2 A3)	A3 = (A1 == A2) ? 1 : 0
NE	(A1 A2 -- A1 A2 A3)	A3 = (A1 != A2) ? 1 : 0
NOT	(A1 -- A1)	A1 = (A1 > 0) ? 1 : 0
SEL	(A1 A2 A3 -- A1 A2 A3)	A3 = (A3 == 0) ? A1 : A2
GRAPHICS		
M	(C1 C2 --)	MOVE TO (C1,C2)
D	(C1 C2 --)	DRAW TO (C1,C2)
C	(C1 C2 C3 --)	DRAW ARC, CENTER C1,C2 RADIUS C3.
MA	(C1 C2 C3 C4 --)	DRAW MARKER, CODE NUMBER C1, CENTER CURRENT POSITION.
BOX	(C1 C2 C3 C4 --)	DRAW RECTANGLE, BOTTOM LEFT C1,C2, WIDTH C3 HEIGHT C4.
DRSH	(A1 -- A1)	DRAW RELATIVE (0,1) IF A1(1) 0 ELSE MOVE
VDSH	(A1 -- A1)	DRAW POLYLINE X=A1,Y=A2, OPEN
PTH	(A1 -- A1)	DRAW POLYLINE X=A1,Y=A2, CLOSED
CRVO	(A1 A2 --)	DRAW SMOOTH CURVE X=A1,Y=A2, OPEN
CRVC	(A1 A2 --)	DRAW SMOOTH CURVE X=A1,Y=A2, CLOSED
BEZO	(A1 A2 --)	DRAW BEZIER CURVE, CONTROL POLYGON X=A1,Y=A2, OPEN
BEZC	(A1 A2 --)	DRAW BEZIER CURVE, CONTROL POLYGON X=A1,Y=A2, CLOSED
TVF	(C1 C2 -- C1)	DRAW CONTENTS OF STRING REGISTER C1 AS TEXT.
TVI	(C1 C2 -- C1)	DRAW INT(C1) AS TEXT USING FORMAT STRING IN C2.
TS	(C1 --)	DRAW SYMBOL WITH CODE NUMBER C1
TM	(C1 --)	DRAW MARKER WITH CODE NUMBER C1
TC	(C1 --)	DRAW CHARACTER WITH CODE C1 FROM CURRENT ALPHABET
TH	(C1 --)	SET TEXT/MARKER/SYMBOL HEIGHT
TA	(C1 --)	SET TEXT DRAWING ANGLE IN DEGREES CCW
TSC	(C1 --)	START TEXT CONTINUATION
TEF	(C1 --)	END TEXT CONTINUATION
TELEN	(C1 -- C1)	GET THE LENGTH IN BOUNDS AT UNIT HEIGHT OF STRING IN REGISTER C1.
ROT	(A1 A2 C3 -- A1 A2)	ROTATE X=A1,Y=A2 BY C3 RADIANS ABOUT ORIGIN.
TRN	(A1 A2 C3 C4 -- A1 A2)	TRANSLATE X=A1,Y=A2 BY C3,C4
SCALE	(A1 A2 C3 C4 -- A1 A2)	SCALE X=A1,Y=A2 BY C3,C4
LAB	(C1 C2 C3 C4 C5 --)	LABEL AT (C1,C2), LEN C3, ANG C4, STRING REG C5
NUMBER TYPE CONVERSIONS		
INT	(A1 -- A1)	TRUNCATION A1 = INT(A1)
FRAC	(A1 -- A1)	FRACTIONAL PART A1 = FRAC(A1)
FLR	(A1 -- A1)	A1 = FLOOR(A1)
CEIL	(A1 -- A1)	A1 = CEILING(A1)
SYSTEM		
STO or =	(A1 C2 -- A1)	STORE A1(I) (A.K.A. C1) IN REGISTER C2
RCL or (HASH)	(C1 -- C1)	RECALL STACK VALUE FROM REGISTER INT(C1)
P	(A1 C2 C3 -- A1)	PRINT TOS ARRAY FIRST AND LAST ELEMENTS
PE	(A1 C2 C3 -- A1)	PRINT ELEMENTS C2 TO C3 OF A1 IN FREE FORMAT
PC	(C1 -- C1)	PRINT C1 (TOS A1(I)) IN FREE FORMAT
DUMP	(--)	PRINT ALL STACK LEVELS FIRST AND LAST ELEMENTS
R2D	(A1 -- A1)	CONVERT RADIANS TO DEGREES
D2R	(A1 -- A1)	CONVERT DEGREES TO RADIANS

